

Chapter 3: The Cox Regression Model

Modelling Survival Data in Medical Research

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Part 3

Chapter 3 Outline

1. Interpreting Parameters
2. Estimating $\hat{S}(t)$ and $\hat{H}(t)$
3. Risk-Adjusted Survivor Function
4. Concordance, R^2 , ROC Curves
5. Cox Model and the Log-Rank Test
6. Looking Ahead
7. Q&A

Interpretation: The Key Principle

Every coefficient in the Cox model is a log-hazard ratio

$$\hat{\beta}_j = \log \frac{h(t \mid x_j = x + 1, \text{ others fixed})}{h(t \mid x_j = x, \text{ others fixed})}$$

Three golden rules

- 1 $e^{\hat{\beta}_j}$: HR for 1-unit increase in X_j , *all other variables held constant*.
- 2 The HR does **not** depend on t — constant over time (proportional hazards).
- 3 The HR does **not** depend on the values of other variables (additive log-hazard model).

A HR of 2.0 means: at any time t , the individual with x_j one unit higher faces *twice* the instantaneous risk of death, holding all other variables constant.

Interpreting a Variate: Detailed Example

Blood urea nitrogen (Bun) in the myeloma model

$$\hat{\beta}_{\text{Bun}} = 0.021, \text{ se} = 0.006, P = 0.001.$$

Estimated hazard ratio per 1 unit:

$$e^{0.021} = 1.021 \quad (+2.1\% \text{ hazard})$$

Per 10 units: $e^{0.21} = 1.23$ (+23%)

Per 20 units: $e^{0.42} = 1.52$ (+52%)

95% CI for HR per 10 units:

$$\begin{aligned} & 10 \times \hat{\beta} \pm 1.96 \times 10 \times \text{se} \\ &= 0.21 \pm 1.96(0.06) = (0.09, 0.33) \\ &\Rightarrow \text{HR} \in (e^{0.09}, e^{0.33}) = (1.09, 1.39) \end{aligned}$$

Interpretation:

After adjusting for serum haemoglobin, every 10-unit increase in blood urea nitrogen is associated with a 23% increase in the hazard of death from multiple myeloma.

Remember: linearity assumption

This same 23% increase applies whether Bun goes from 10 to 20, or from 50 to 60. We should verify this using the non-linearity test.

Interpreting Factor Coefficients

Example 3.10 — Hypernephroma: nephrectomy patients, 3 age groups

Model: $h_j(t) = e^{\alpha_j} h_0(t)$, with $\alpha_1 = 0$ (age < 60 = baseline).

Level	$\hat{\alpha}$	se	HR
< 60 (baseline)	0	—	1.00
60–70	−0.065	0.498	0.94
> 70	1.824	0.682	6.20

95% CI for HR (> 70 vs < 60):

$$e^{1.824 \pm 1.96(0.682)} = (1.63, 23.59)$$

Does not include 1 $\Rightarrow$ significant.

Non-baseline comparison:

HR for > 70 vs. 60–70:

$$\hat{\alpha}_3 - \hat{\alpha}_2 = 1.889, \quad e^{1.889} = 6.61$$

SE from variance-covariance matrix:

$$\text{var}(\hat{\alpha}_3 - \hat{\alpha}_2) = 0.465 + 0.248 - 2(0.083) = 0.547$$

se = 0.740; 95% CI for HR: (1.55, 28.18).

Tip: redefine the baseline level to get any comparison directly from R output.

Adjusted vs. Unadjusted Hazard Ratios

Example 3.12 — Prostatic cancer: treatment (DES) adjusted for tumour size and Gleason index

	$\hat{\beta}$	se	HR
Treatment (unadjusted, only Treat in model)	-1.978	—	0.14
Treatment (adjusted for Size + Index)	-1.113	1.203	0.33

Why do the HRs differ?

Patients allocated to DES (active treatment) had on average larger, more aggressive tumours. Without adjustment, the treatment looks *even more beneficial* because the comparison is confounded by tumour characteristics.

Adjusted interpretation:

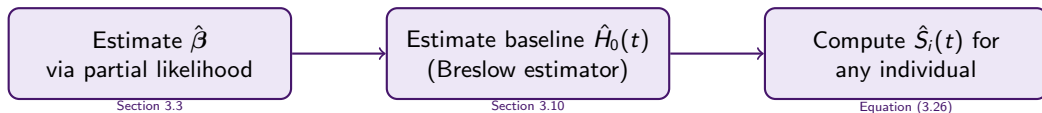
After adjusting for tumour size and Gleason index, DES patients have $\approx 33\%$ of the hazard of placebo patients (unadjusted: 14%).

$P = 0.327$: not significant — no evidence of a treatment effect after proper adjustment.

Going from $\hat{\beta}$ to $\hat{S}_i(t)$: The Big Picture

So far we have focused on estimating β (the covariate effects).

Now we want to **predict** the survival function for a specific patient with covariate vector $\mathbf{x}_i$.



Key formula

$$\hat{S}_i(t) = \hat{S}_0(t)^{\exp(\hat{\beta}' \mathbf{x}_i)}, \quad \hat{H}_i(t) = e^{\hat{\beta}' \mathbf{x}_i} \hat{H}_0(t)$$

When there are no covariates, $\hat{S}_0(t)$ reduces to the **Kaplan-Meier** estimate.
The Cox model generalises Chapter 2 completely.

The Breslow Estimator of $\hat{H}_0(t)$

Breslow estimator (widely used approximation)

$$\tilde{H}_0(t) = \sum_{j: t_{(j)} \leq t} \frac{d_j}{\sum_{I \in R(t_{(j)})} e^{\hat{\beta}' \mathbf{x}_I}}$$

d_j = number of deaths at $t_{(j)}$, denominator = sum of risk scores in the risk set.

Special case (no covariates):

$e^{\hat{\beta}' \mathbf{x}_I} = 1$, so denominator = n_j .

$$\tilde{H}_0(t) = \sum_{j: t_{(j)} \leq t} \frac{d_j}{n_j} = \text{Nelson-Aalen!}$$

Estimated survivor function for individual i :

$$\hat{S}_i(t) = \exp\left(-e^{\hat{\eta}_i} \tilde{H}_0(t)\right)$$

where $\hat{\eta}_i = \hat{\beta}' \mathbf{x}_i$ is the risk score.

In R:

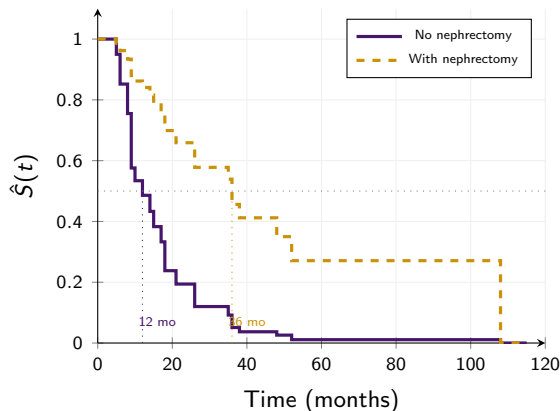
```
survfit(cox.fit, newdata =  
data.frame(...))
```


Example 3.14 — Hypernephroma: Plotting Survivor Functions

Model: $\hat{h}_i(t) = e^{0.013A_{2i} + 1.342A_{3i} - 1.412N_i} \hat{h}_0(t)$

Key results from the plot:

- ▶ Nephrectomy gives a **large survival benefit** for the < 60 age group
- ▶ Estimated median without nephrectomy: 12 months
- ▶ Estimated median with nephrectomy: 36 months
- ▶ PH assumption: the two curves **never cross**
- ▶ $\hat{S}_i(t) = \hat{S}_0(t)^{e^{-1.412}} = \hat{S}_0(t)^{0.244}$
(survival is “boosted” by nephrectomy)



The Risk-Adjusted Survivor Function: Motivation

After fitting the Cox model, we have **individual survivor functions** $\hat{S}_i(t)$ for each of n patients.

Question

How do we summarise the survival experience for a group of patients, after adjusting for their covariate distribution?

Answer:

Risk-adjusted survivor function

$$\hat{\hat{S}}(t) = \frac{1}{n} \sum_{i=1}^n \hat{S}_i(t) = \frac{1}{n} \sum_{i=1}^n \hat{S}_0(t)^{e^{\hat{\eta}_i}}$$

Why not just plot the KM curve?

In an observational study, the two groups being compared may differ in their covariate distributions (e.g. older patients in one group, larger tumours in another).

The unadjusted KM curve conflates the *treatment/group effect* with the *covariate effect*.

The risk-adjusted curve **removes** the covariate effect, showing a fairer comparison.

Example 3.16 — Prostatic Cancer: The Impact of Adjustment

Stratified Cox model: separate baseline hazards for DES and placebo groups, adjusted for Size and Index.

Key observation from the plot:

After risk adjustment, the **treatment difference shrinks** compared to the raw KM curves.

Why?

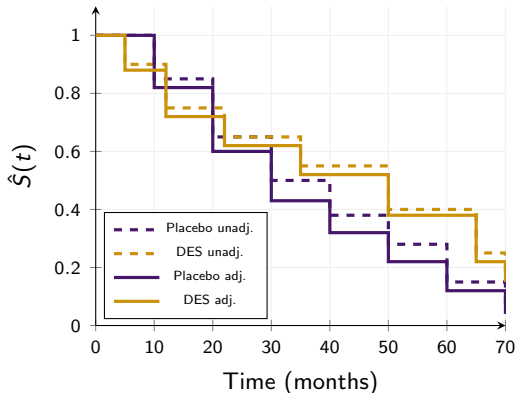
Patients on DES happened to have more aggressive tumours (larger, higher Gleason index). The raw KM comparison was confounded.

Unadjusted HR: 0.14

Adjusted HR: 0.33

Adjustment reduces apparent treatment benefit.

Adjusted (solid) vs. unadjusted KM (dashed)



How Good Is the Model? Three Questions

After fitting the Cox model, we want to ask:

Discrimination

How well does the model *separate* patients with long vs. short survival?

⇒ c-statistic, D-statistic

Explained variation

What *fraction* of variability in survival times is explained?

⇒ R_P^2 , R_D^2 , R_W^2

Temporal prediction

Does the model predict short-term death differently from long-term?

⇒ Time-dependent ROC, $AUC(t)$

High discrimination and high R^2 together indicate a useful model.
But even the best models in survival analysis typically have $R^2 \approx 0.3$ – 0.5 .

Concordance: The c-Statistic

Harrell's c-statistic: the probability that, for a randomly chosen pair of individuals, the one with *shorter survival* has the *higher risk score*.

How to compute:

- 1 Consider all usable pairs: both died, or one died before the other's censoring time
- 2 A pair is **concordant** if the individual with higher risk score died first
- 3 $c\text{-statistic} = (\text{number concordant}) / (\text{number usable pairs})$

Interpretation

- ▶ $c = 0.5$: random, no predictive ability
- ▶ $c = 0.7$: reasonable discrimination
- ▶ $c = 1.0$: perfect prediction

Limitation of the c-statistic:

Its value depends on the censoring pattern (pairs with censored survival times are excluded).

Gonen-Heller measure:

A concordance probability that is *not affected* by censoring:

$$\hat{K} = \frac{2}{n(n-1)} \sum_{i < j} \frac{I(\hat{\eta}_j < \hat{\eta}_i)}{1 + e^{\hat{\eta}_j - \hat{\eta}_i}}$$

Preferred in modern analyses.

R^2 Measures of Explained Variation

R_P^2 (Kent-O'Quigley):

$$R_P^2 = \frac{\hat{V}_P}{\hat{V}_P + \pi^2/6}$$

$\hat{V}_P$ = variance of risk scores $\hat{\eta}_i$. The $\pi^2/6$ term comes from the logistic distribution (limiting distribution under PH).

R_D^2 (Royston-Sauerbrei):

$$R_D^2 = \frac{D_0^2}{D_0^2 + \pi^2/6}$$

Based on D-statistic (separability of risk scores).

Multiple myeloma: Hb + Bun model

Measure	Selected	Full
c-statistic	0.67	0.71
R_P^2	0.27	0.31
R_D^2	0.20	0.27
R_W^2	0.30	0.34

The selected model (2 variables) explains about **27–30%** of the variation in survival times.

Adding 5 more variables improves this only modestly.

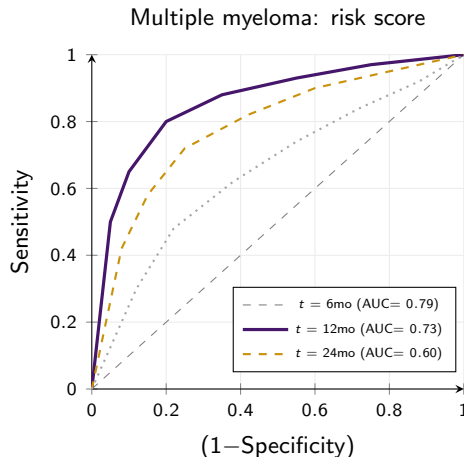
Time-Dependent ROC Curves: Motivation

The c-statistic and R^2 give **overall** summaries. But:

- ▶ A model might predict 6-month survival very well but fail at predicting 5-year survival
- ▶ Different biomarkers may be better at different time horizons
- ▶ Clinical decisions often depend on a specific time window

Solution: construct an ROC curve at each time point t of interest, using the risk score as the biomarker.

Disease status at time t : $D(t) = 1$ if $T < t$ (died before t), $D(t) = 0$ if $T \geq t$ (alive at t).



Reading the Time-Dependent ROC Curves

From the previous plot (multiple myeloma):

Time	AUC
6 months	0.79
12 months	0.73
18 months	0.69
24 months	0.60

Conclusion:

The risk score (based on Hb and Bun) is a better *short-term* predictor than a long-term predictor. As time increases, other factors dominate and the model loses predictive power.

What does $AUC = 0.79$ at 6 months mean?

If we pick two random patients — one who dies within 6 months and one who survives beyond 6 months — the risk score correctly identifies which is which 79% of the time.

AUC guide

0.5: random	0.7: adequate
0.8: good	0.9: excellent

The Deep Connection Between Cox and Log-Rank

Testing $H_0 : \beta = 0$ in the Cox model via a **score test** gives *exactly* the log-rank statistic (no ties case):

Score test at $\beta = 0$:

$$u(0) = \sum_{j=1}^r \left(d_{1j} - \frac{n_{1j}}{n_j} \right) = U_L$$

$$i(0) = \sum_{j=1}^r \frac{n_{1j}n_{2j}}{n_j^2} = V_L$$

$$u(0)^2 / i(0) = U_L^2 / V_L = W_L \text{ (log-rank!)}$$

What this means:

- ▶ The log-rank test is the *optimal* test under proportional hazards
- ▶ It is the score test of the Cox model
- ▶ The Cox model *generalises* the log-rank test to continuous covariates and multiple groups
- ▶ The Cox model also gives a direct estimate of the hazard ratio (e^β), which the log-rank test does not

Comparing the Two Approaches: When to Use Which

	Log-Rank Test (Ch. 2)	Cox Model (Ch. 3)
Purpose	Test for group differences	Estimate effects + test
Number of groups	2 (or several)	Unlimited
Covariates	Cannot adjust	Adjusts simultaneously
Output	<i>P</i> -value only	<i>P</i> -value + HR + 95% CI
Continuous variables	Requires grouping	Handles directly
Baseline hazard	Not estimated	Can be estimated
Assumptions	Same as Cox	PH + linearity

The Cox model is strictly more powerful and informative than the log-rank test. Use the log-rank for quick two-group comparisons; the Cox model for everything else.

Chapter 3 Summary

The Cox model: $h_i(t) = e^{\beta' x_i} h_0(t)$

- ▶ $h_0(t)$ unspecified (semi-parametric)
- ▶ β via partial likelihood
- ▶ Continuous + categorical + interactions
- ▶ Generalises the log-rank test

Model building:

- ▶ Nested models: $-2 \log \hat{L}$ (LRT)
- ▶ Non-nested: AIC, BIC
- ▶ 4-step manual strategy
- ▶ Lasso for high-dimensional problems
- ▶ Fractional polynomials for non-linearity

Interpretation:

- ▶ $e^{\hat{\beta}_j}$: HR per unit increase in X_j , adjusted for other terms
- ▶ Factors: indicator variables, exponentiate

After fitting:

- ▶ $\hat{S}_i(t) = \hat{S}_0(t)^{e^{\hat{\eta}_i}}$ (Breslow)
- ▶ Risk-adjusted survivor function

Assessment:

- ▶ c-statistic, Gonen-Heller $\hat{K}$
- ▶ R_P^2 , R_D^2 , R_W^2
- ▶ Time-dependent AUC(t)

Looking Ahead

Chapter 4: Model checking. *Chapters 5–6:* Parametric $h_0(t)$.

Questions?

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